

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

Name: Suvedhan G	Register No: 192524381
Department: CSE - AIDS	Course Code: DSA0604
Course Name: Data Handling and Visualization	

DATA VISUALIZATION — SET 6 TO 10

SET 6: PRODUCT SALES ANALYSIS

Dataset: Monthly Product Sales

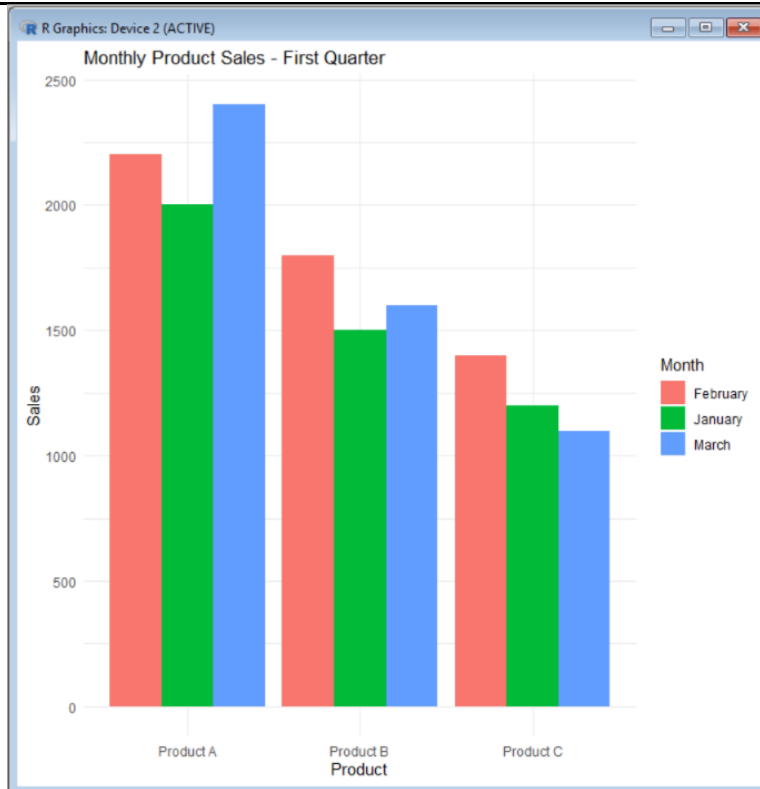
Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Q1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

CODE:

```
library(ggplot2)
library(tidyr)
sales_data <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)
sales_long <- pivot_longer(
  sales_data,
  cols = c(January, February, March),
  names_to = "Month",
  values_to = "Sales"
)
ggplot(sales_long, aes(x = Product, y = Sales, fill = Month)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Monthly Product Sales - First Quarter",
    x = "Product",
    y = "Sales"
  ) +
  theme_minimal()
```

OUTPUT:

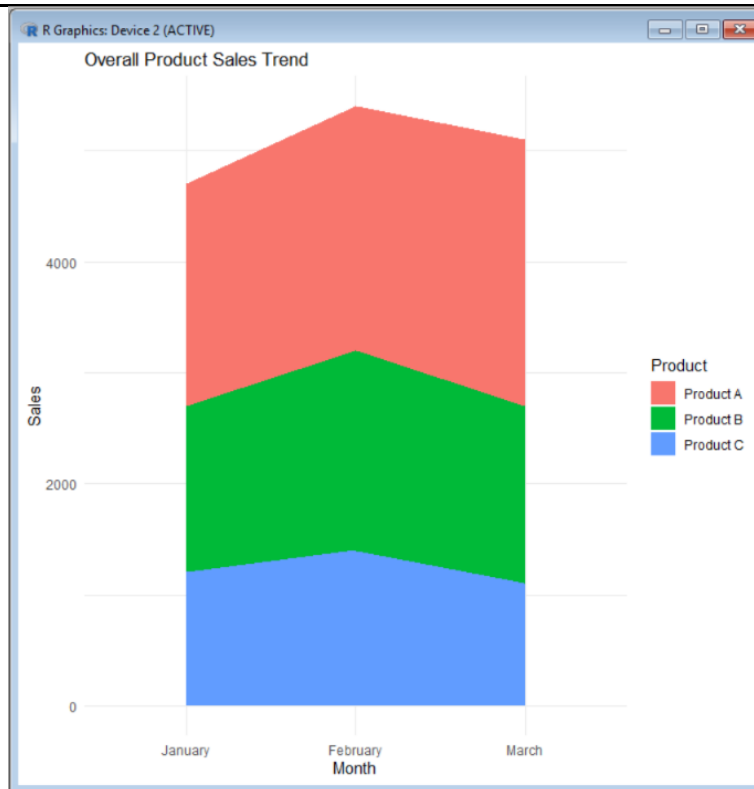


Q2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

CODE:

```
library(ggplot2)
library(tidyr)
sales_data <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)
sales_long <- pivot_longer(
  sales_data,
  cols = c(January, February, March),
  names_to = "Month",
  values_to = "Sales"
)
sales_long$Month <- factor(
  sales_long$Month,
  levels = c("January", "February", "March")
)
ggplot(sales_long, aes(x = Month, y = Sales, fill = Product, group =
  Product)) +
  geom_area() +
  labs(
    title = "Overall Product Sales Trend",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()
```

OUTPUT:



Q3. Build a table to show the monthly sales data for each product. Label the table elements.

CODE:

```
sales_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  January_Sales = c(2000, 1500, 1200),
  February_Sales = c(2200, 1800, 1400),
  March_Sales = c(2400, 1600, 1100)
)
print(sales_data)
```

OUTPUT:

```
> sales_data <- data.frame(
+   Product_ID = c(1, 2, 3),
+   Product_Name = c("Product A", "Product B", "Product C"),
+   January_Sales = c(2000, 1500, 1200),
+   February_Sales = c(2200, 1800, 1400),
+   March_Sales = c(2400, 1600, 1100)
+ )
>
> print(sales_data)
  Product_ID Product_Name January_Sales February_Sales March_Sales
1          1   Product A         2000          2200         2400
2          2   Product B         1500          1800         1600
3          3   Product C         1200          1400         1100
> |
```

Q4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

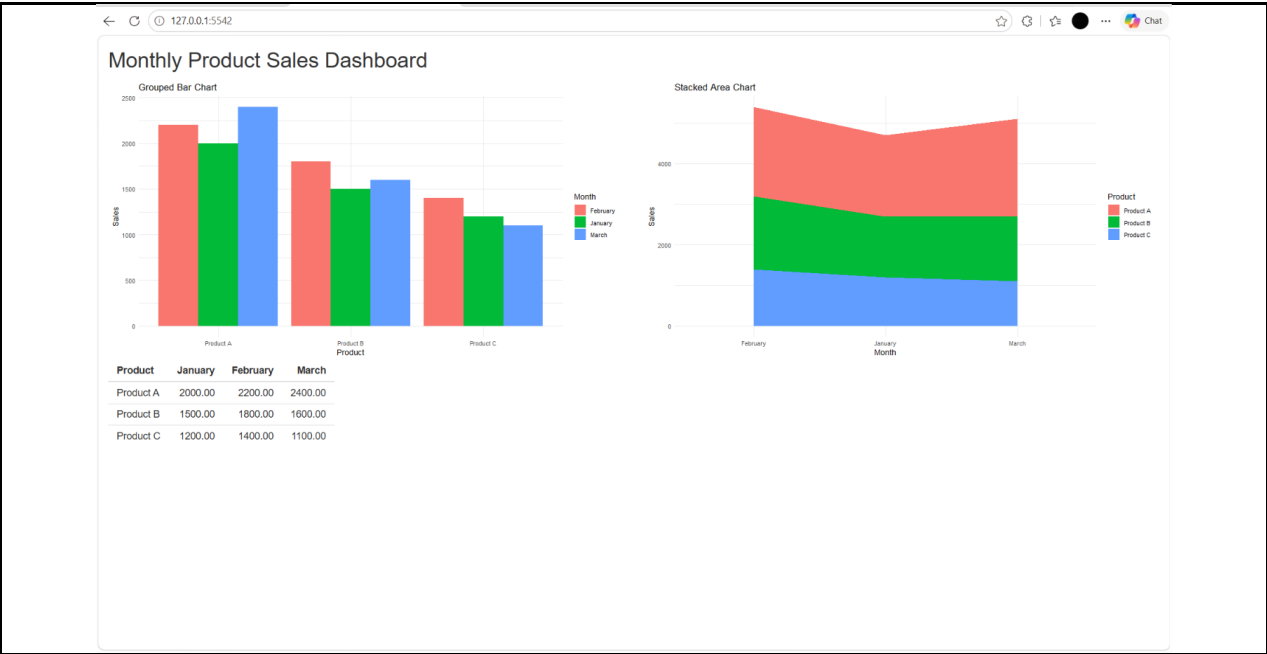
CODE:

```

library(shiny)
library(ggplot2)
library(tidyr)
sales_data <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)
sales_long <- pivot_longer(
  sales_data,
  cols = c(January, February, March),
  names_to = "Month",
  values_to = "Sales"
)
ui <- fluidPage(
  titlePanel("Monthly Product Sales Dashboard"),
  fluidRow(
    column(6, plotOutput("groupBar")),
    column(6, plotOutput("areaChart"))
  ),
  fluidRow(
    column(12, tableOutput("salesTable"))
  )
)
server <- function(input, output) {
  output$groupBar <- renderPlot({
    ggplot(sales_long, aes(Product, Sales, fill = Month)) +
    geom_bar(stat = "identity", position = "dodge") +
    labs(
      title = "Grouped Bar Chart",
      x = "Product",
      y = "Sales"
    ) +
    theme_minimal()
  })
  output$areaChart <- renderPlot({
    ggplot(sales_long, aes(Month, Sales, fill = Product, group = Product)) +
    geom_area() +
    labs(
      title = "Stacked Area Chart",
      x = "Month",
      y = "Sales"
    ) +
    theme_minimal()
  })
  output$salesTable <- renderTable({
    sales_data
  })
}
shinyApp(ui, server)

```

OUTPUT:



SET 7: CUSTOMER DEMOGRAPHICS ANALYSIS

Dataset: Customer Demographics

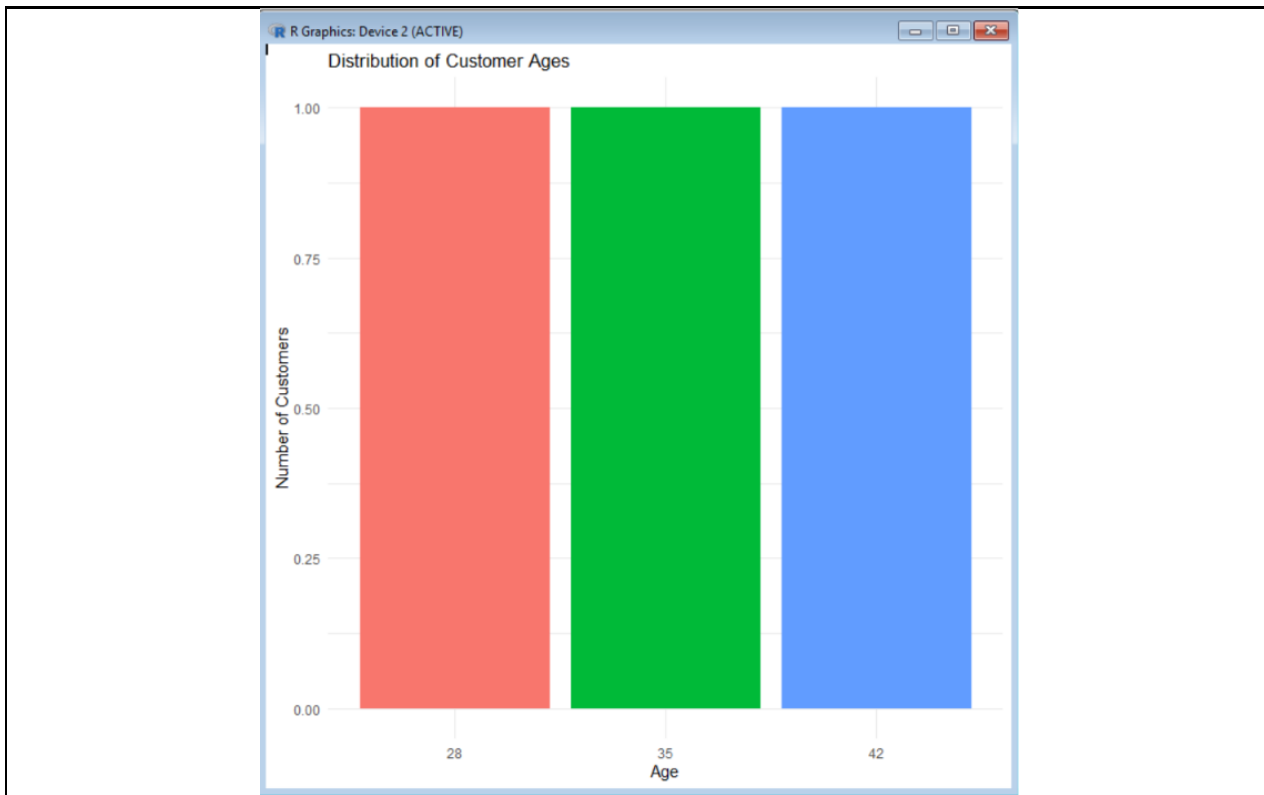
Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Q1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

CODE:

```
library(ggplot2)
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)
ggplot(customer_data, aes(x = factor(Age), fill = factor(Age))) +
  geom_bar() +
  labs(
    title = "Distribution of Customer Ages",
    x = "Age",
    y = "Number of Customers"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

OUTPUT:

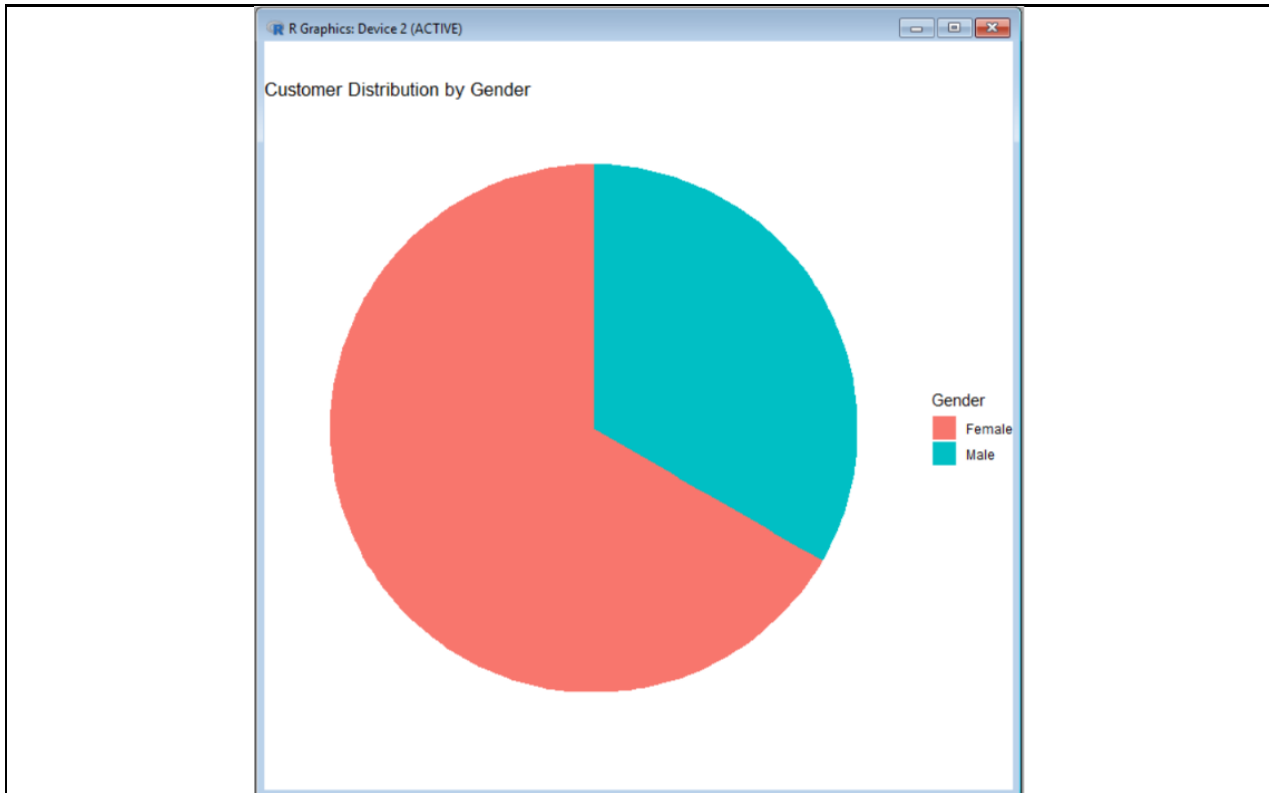


Q2. Generate a pie chart to represent the distribution of customers by gender.

CODE:

```
library(ggplot2)
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)
gender_data <- as.data.frame(table(customer_data$Gender))
colnames(gender_data) <- c("Gender", "Count")
ggplot(gender_data, aes(x = "", y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  labs(
    title = "Customer Distribution by Gender",
    fill = "Gender"
  ) +
  theme_void()
```

OUTPUT:



Q3. Build a table to show the demographic information for each customer. Label the table elements.

CODE:

```
customer_data <- data.frame(  
  Customer_ID = c(1, 2, 3),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),  
  Income = c(50000, 60000, 75000)  
)  
print(customer_data)
```

OUTPUT:

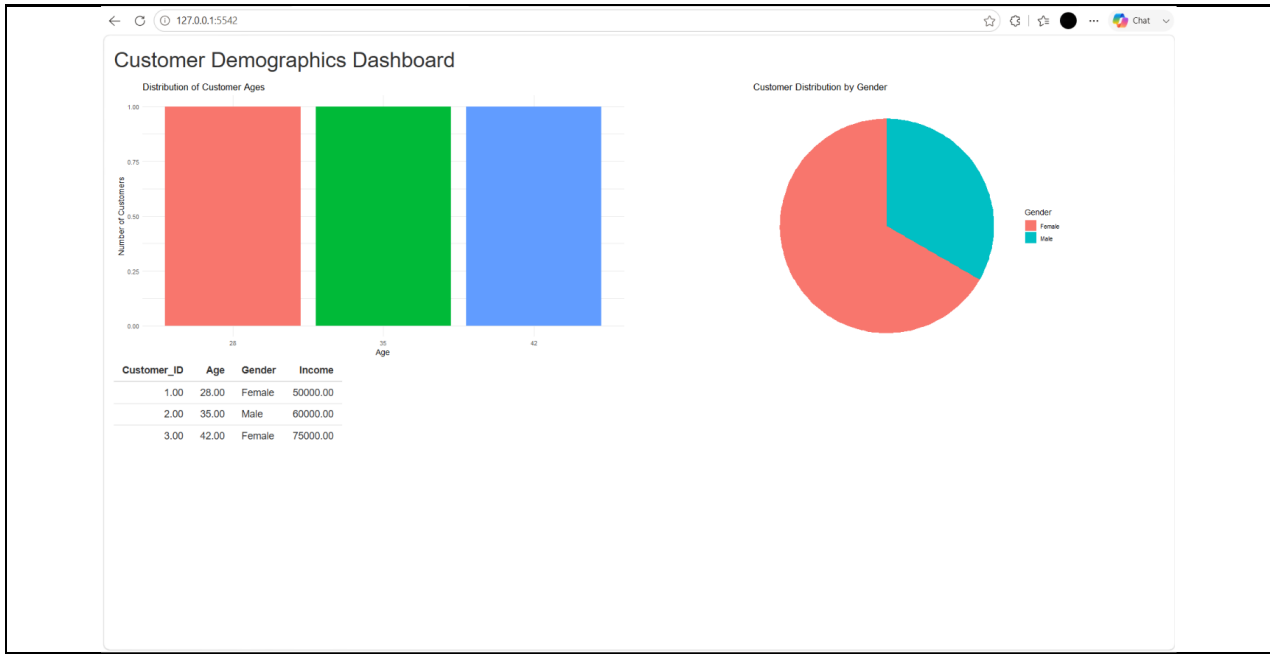
Customer_ID	Age	Gender	Income
1.00	28.00	Female	50000.00
2.00	35.00	Male	60000.00
3.00	42.00	Female	75000.00

Q4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

CODE:

```
library(ggplot2)
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)
gender_data <- as.data.frame(table(customer_data$Gender))
colnames(gender_data) <- c("Gender", "Count")
ui <- fluidPage(
  titlePanel("Customer Demographics Dashboard"),
  fluidRow(
    column(6, plotOutput("barPlot")),
    column(6, plotOutput("piePlot"))
  ),
  fluidRow(
    column(12, tableOutput("customerTable"))
  )
)
server <- function(input, output) {
  output$barPlot <- renderPlot({
    ggplot(customer_data, aes(x = factor(Age), fill = factor(Age))) +
      geom_bar() +
      labs(
        title = "Distribution of Customer Ages",
        x = "Age",
        y = "Number of Customers"
      ) +
      theme_minimal() +
      theme(legend.position = "none")
  })
  output$piePlot <- renderPlot({
    ggplot(gender_data, aes(x = "", y = Count, fill = Gender)) +
      geom_bar(stat = "identity", width = 1) +
      coord_polar("y", start = 0) +
      labs(
        title = "Customer Distribution by Gender",
        fill = "Gender"
      ) +
      theme_void()
  })
  output$customerTable <- renderTable({
    customer_data
  })
}
shinyApp(ui, server)
```

OUTPUT:



SET 8: EMPLOYEE PERFORMANCE ANALYSIS

Dataset: Employee Performance

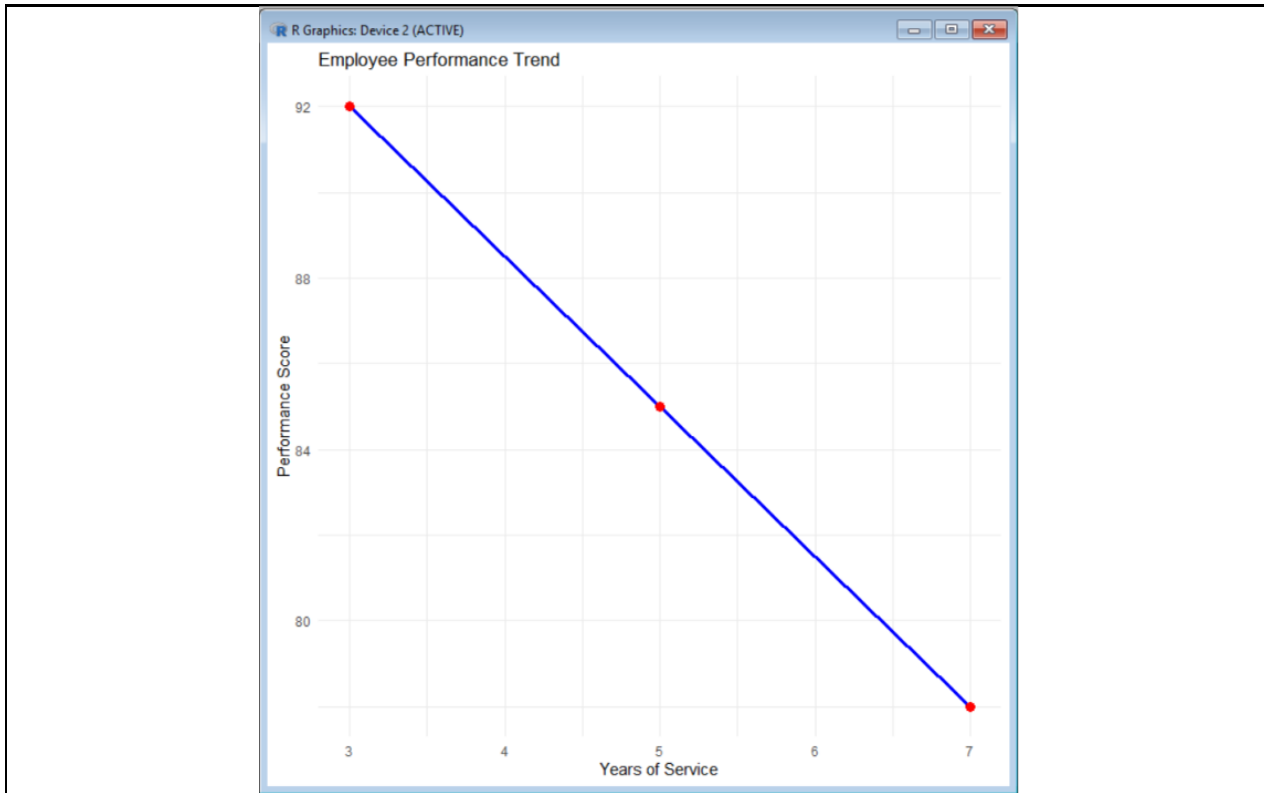
Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Q1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

CODE:

```
library(ggplot2)
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
ggplot(employee_data, aes(x = Years_of_Service, y = Performance_Score,
group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score"
  ) +
  theme_minimal()
```

OUTPUT:

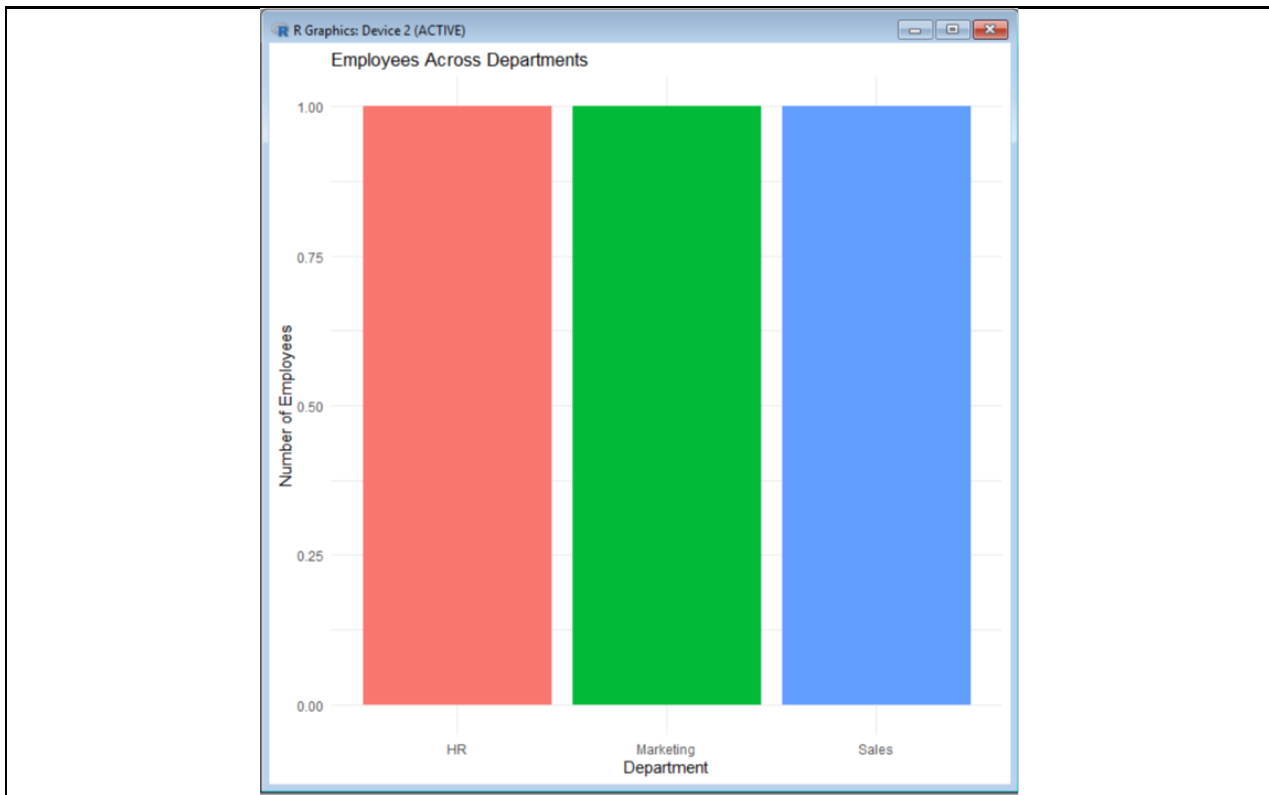


Q2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

CODE:

```
library(ggplot2)
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
ggplot(employee_data, aes(x = Department, fill = Department)) +
  geom_bar() +
  labs(
    title = "Employees Across Departments",
    x = "Department",
    y = "Number of Employees"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

OUTPUT:



Q3. Build a table to display the performance data for each employee. Label the table elements.

CODE:

```
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
print(employee_data)
```

OUTPUT:

Employee_ID	Department	Years_of_Service	Performance_Score
1.00	Sales	5.00	85.00
2.00	HR	3.00	92.00
3.00	Marketing	7.00	78.00

Q4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

CODE:

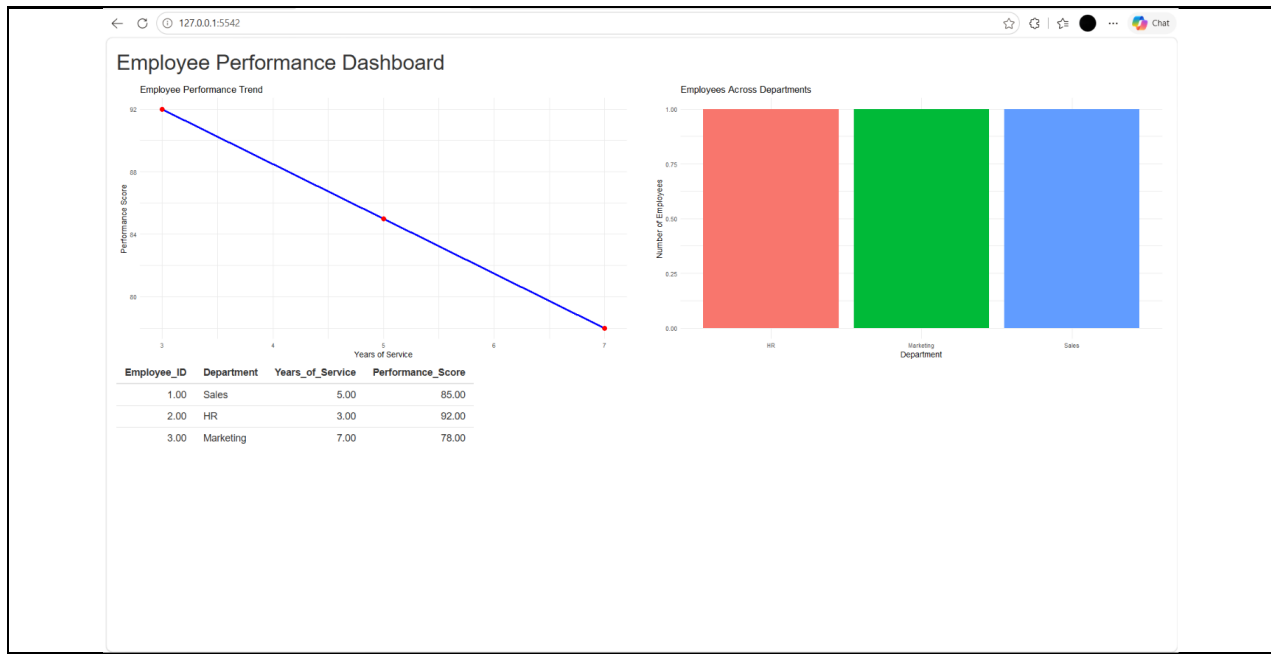
```
library(shiny)
library(ggplot2)
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
```

```

Years_of_Service = c(5, 3, 7),
Performance_Score = c(85, 92, 78)
)
ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
  fluidRow(
    column(6, plotOutput("linePlot")),
    column(6, plotOutput("barPlot"))
  ),
  fluidRow(
    column(12, tableOutput("employeeTable"))
  )
)
server <- function(input, output) {
  output$linePlot <- renderPlot({
    ggplot(employee_data, aes(x = Years_of_Service, y = Performance_Score,
    group = 1)) +
    geom_line(color = "blue", linewidth = 1.2) +
    geom_point(color = "red", size = 3) +
    labs(
      title = "Employee Performance Trend",
      x = "Years of Service",
      y = "Performance Score"
    ) +
    theme_minimal()
  })
  output$barPlot <- renderPlot({
    ggplot(employee_data, aes(x = Department, fill = Department)) +
    geom_bar() +
    labs(
      title = "Employees Across Departments",
      x = "Department",
      y = "Number of Employees"
    ) +
    theme_minimal() +
    theme(legend.position = "none")
  })
  output$employeeTable <- renderTable({
    employee_data
  })
}
shinyApp(ui, server)

```

OUTPUT:



SET 9: ONLINE LEARNING ACTIVITY ANALYSIS

Dataset: *Online_Learning_Activity.csv*

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

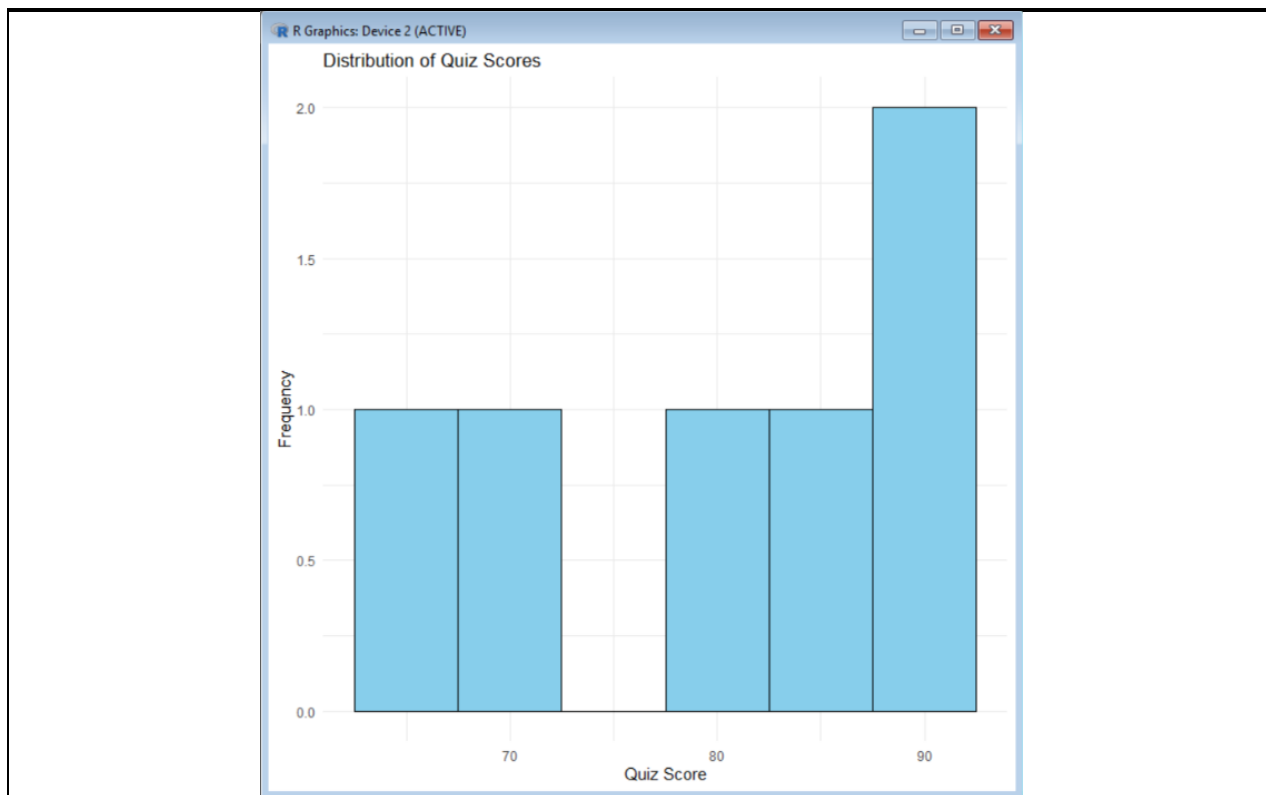
Q1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

CODE:

```
library(ggplot2)
student_data <- data.frame(
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Age = c(20, 22, 19, 21, 23, 20),
  Course = c("R", "R", "SQL", "R", "R", "SQL"),
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),
  Videos_Watched = c(12, 15, 8, 18, 9, 14),
  Quiz_Score = c(78, 85, 65, 92, 70, 88),
  Login_Date =
c("2025-01-05", "2025-01-05", "2025-02-08", "2025-02-08", "2025-03-12", "2025-03-12")
)
ggplot(student_data, aes(x = Quiz_Score)) +
```

```
geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
labs(
  title = "Distribution of Quiz Scores",
  x = "Quiz Score",
  y = "Frequency"
) +
theme_minimal()
ggplot(student_data, aes(x = Course, y = Quiz_Score, fill = Course)) +
geom_boxplot() +
labs(
  title = "Quiz Score by Course",
  x = "Course",
  y = "Quiz Score"
) +
theme_minimal()
```

OUTPUT:



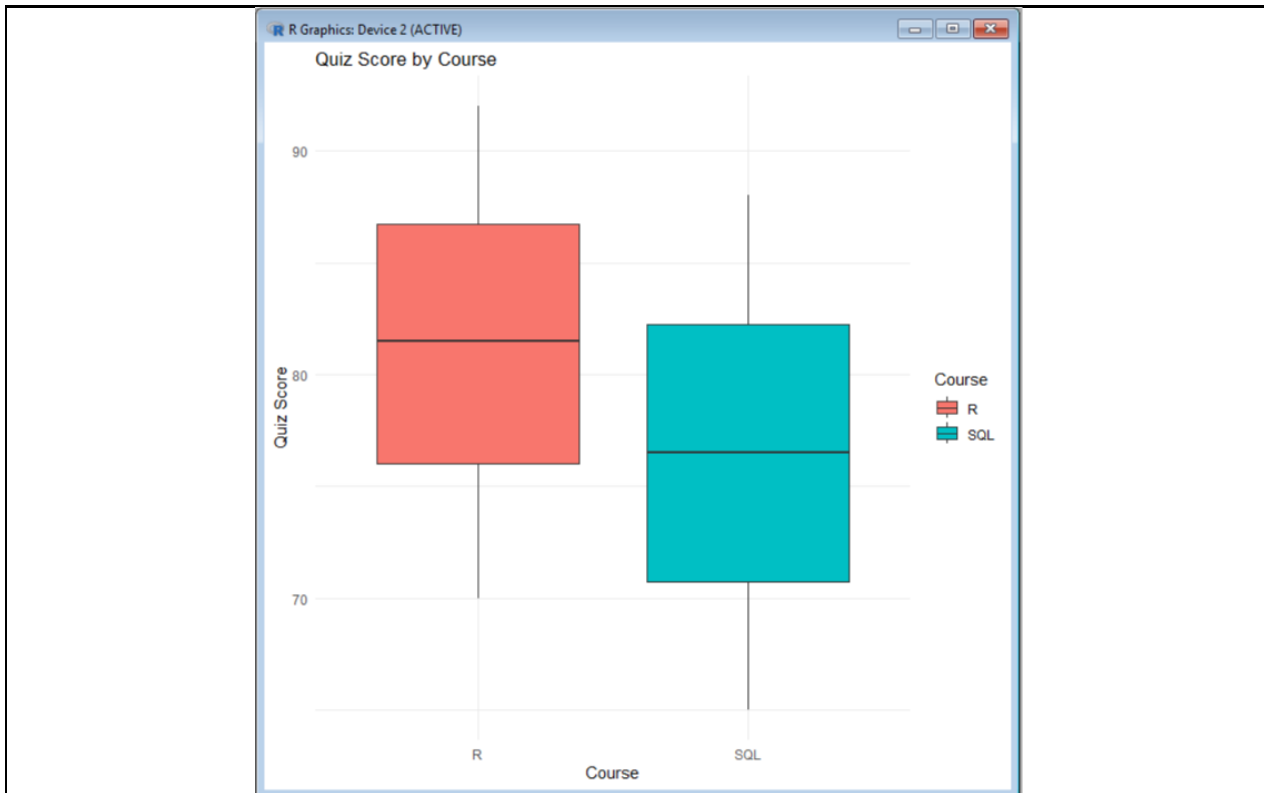
Q2. Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

CODE:

```
library(ggplot2)
student_data <- data.frame(
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Age = c(20, 22, 19, 21, 23, 20),
  Course = c("R", "R", "SQL", "R", "R", "SQL"),
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),
  Videos_Watched = c(12, 15, 8, 18, 9, 14),
  Quiz_Score = c(78, 85, 65, 92, 70, 88)
)
```

```
ggplot(student_data,
  aes(x = Study_Time,
    y = Quiz_Score,
    size = Videos_Watched)) +
  geom_point(alpha = 0.6, color = "blue") +
  labs(
    title = "Study Time vs Quiz Score",
    x = "Study Time (hrs)",
    y = "Quiz Score",
    size = "Videos Watched"
  ) +
  theme_minimal()
```

OUTPUT:



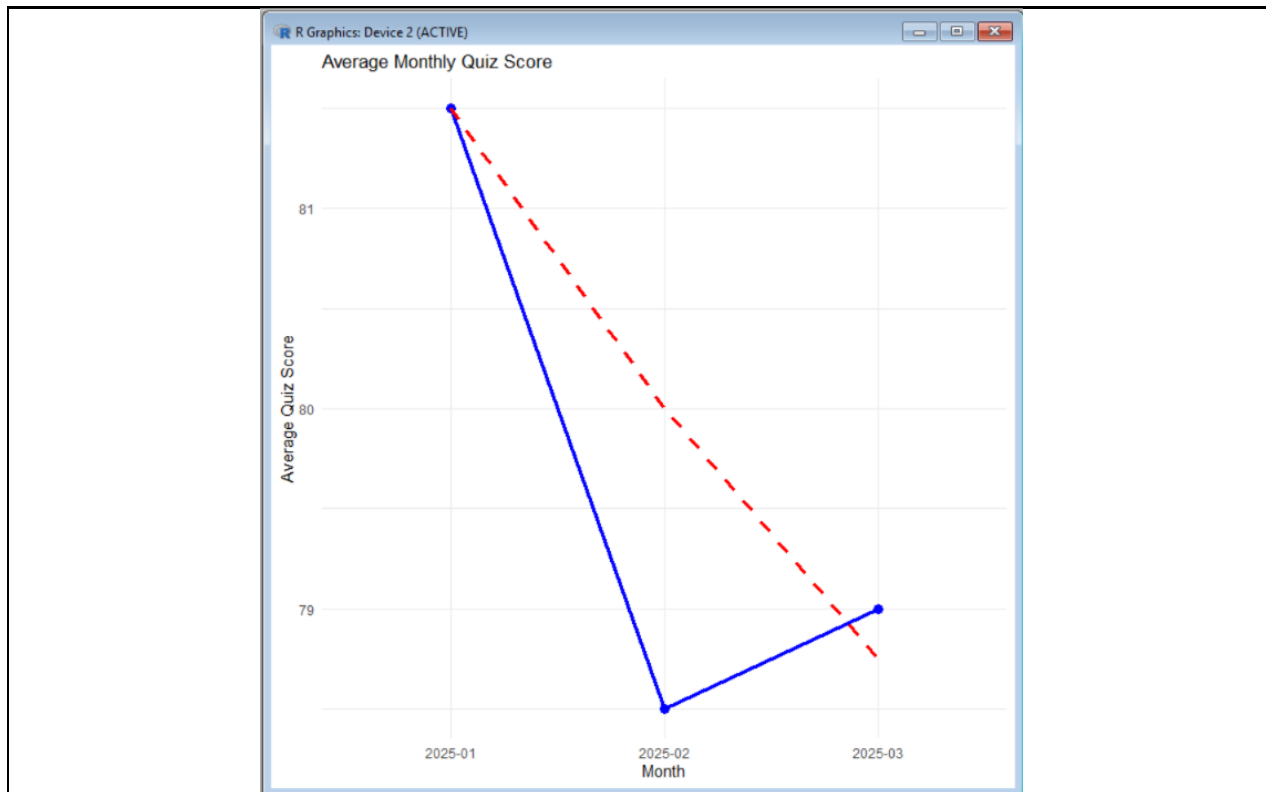
Q3. Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

CODE:

```
library(ggplot2)
student_data <- data.frame(
  Login_Date =
    as.Date(c("2025-01-05", "2025-01-05", "2025-02-08", "2025-02-08", "2025-03-12", "2025-03-12")),
  Quiz_Score = c(78, 85, 65, 92, 70, 88)
)
student_data$Month <- format(student_data$Login_Date, "%Y-%m")
monthly_avg <- aggregate(Quiz_Score ~ Month, student_data, mean)
monthly_avg$Moving_Average <- c(
  monthly_avg$Quiz_Score[1],
  (monthly_avg$Quiz_Score[1] + monthly_avg$Quiz_Score[2]) / 2,
  (monthly_avg$Quiz_Score[2] + monthly_avg$Quiz_Score[3]) / 2
)
```

```
ggplot(monthly_avg, aes(x = Month, y = Quiz_Score, group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(size = 3, color = "blue") +
  geom_line(aes(y = Moving_Average),
    color = "red",
    linetype = "dashed",
    linewidth = 1.2) +
  labs(
    title = "Average Monthly Quiz Score",
    x = "Month",
    y = "Average Quiz Score"
  ) +
  theme_minimal()
```

OUTPUT:



Q4. Import Online_Learning_Activity.csv into Tableau. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

CODE:

```
library(shiny)
library(ggplot2)
student_data <- data.frame(
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Age = c(20, 22, 19, 21, 23, 20),
  Course = c("R", "R", "SQL", "R", "R", "SQL"),
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),
  Videos_Watched = c(12, 15, 8, 18, 9, 14),
  Quiz_Score = c(78, 85, 65, 92, 70, 88)
)
```

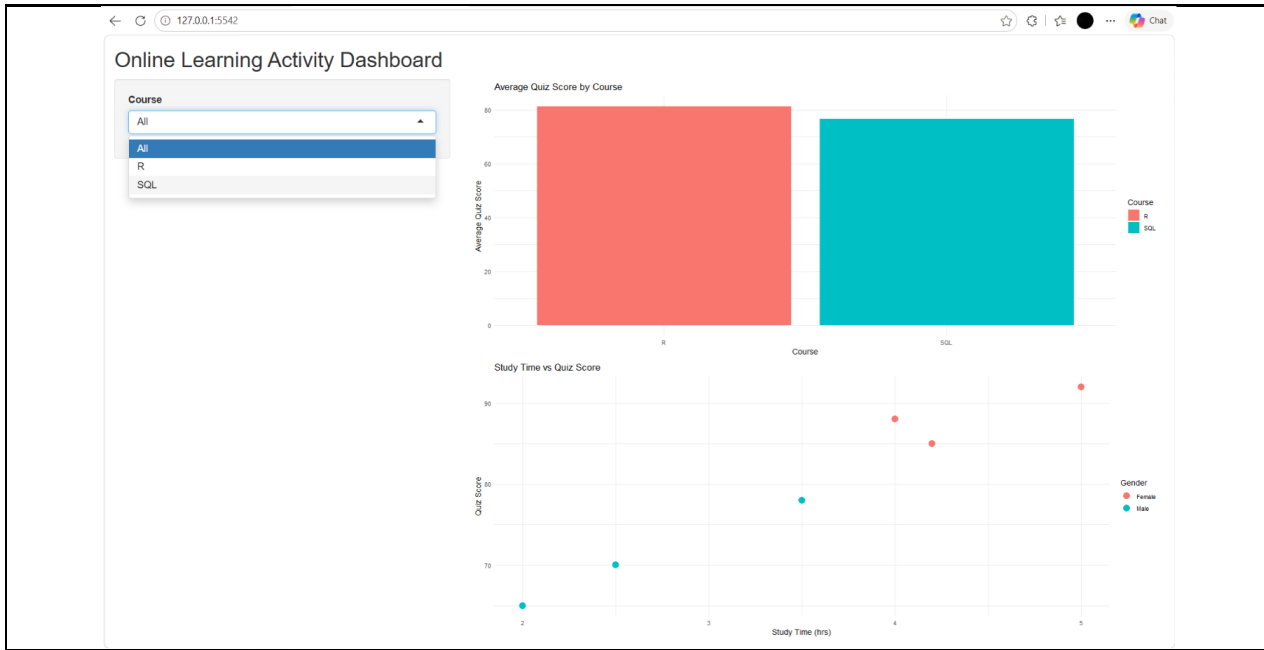


```

ui <- fluidPage(
  titlePanel("Online Learning Activity Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput("course", "Course",
        choices = c("All", unique(student_data$Course)),
        selected = "All")
    ),
    mainPanel(
      plotOutput("barPlot"),
      plotOutput("scatterPlot")
    )
  )
)
server <- function(input, output) {
  filtered <- reactive({
    if (input$course == "All") {
      student_data
    } else {
      subset(student_data, Course == input$course)
    }
  })
  output$barPlot <- renderPlot({
    avg <- aggregate(Quiz_Score ~ Course, filtered(), mean)
    ggplot(avg, aes(x = Course, y = Quiz_Score, fill = Course)) +
      geom_bar(stat = "identity") +
      labs(
        title = "Average Quiz Score by Course",
        x = "Course",
        y = "Average Quiz Score"
      ) +
      theme_minimal()
  })
  output$scatterPlot <- renderPlot({
    ggplot(filtered(),
      aes(x = Study_Time,
        y = Quiz_Score,
        color = Gender)) +
      geom_point(size = 4) +
      labs(
        title = "Study Time vs Quiz Score",
        x = "Study Time (hrs)",
        y = "Quiz Score"
      ) +
      theme_minimal()
  })
}
shinyApp(ui, server)

```

OUTPUT:



SET 10: SURVEY RESPONSES ANALYSIS

Dataset: Survey Responses

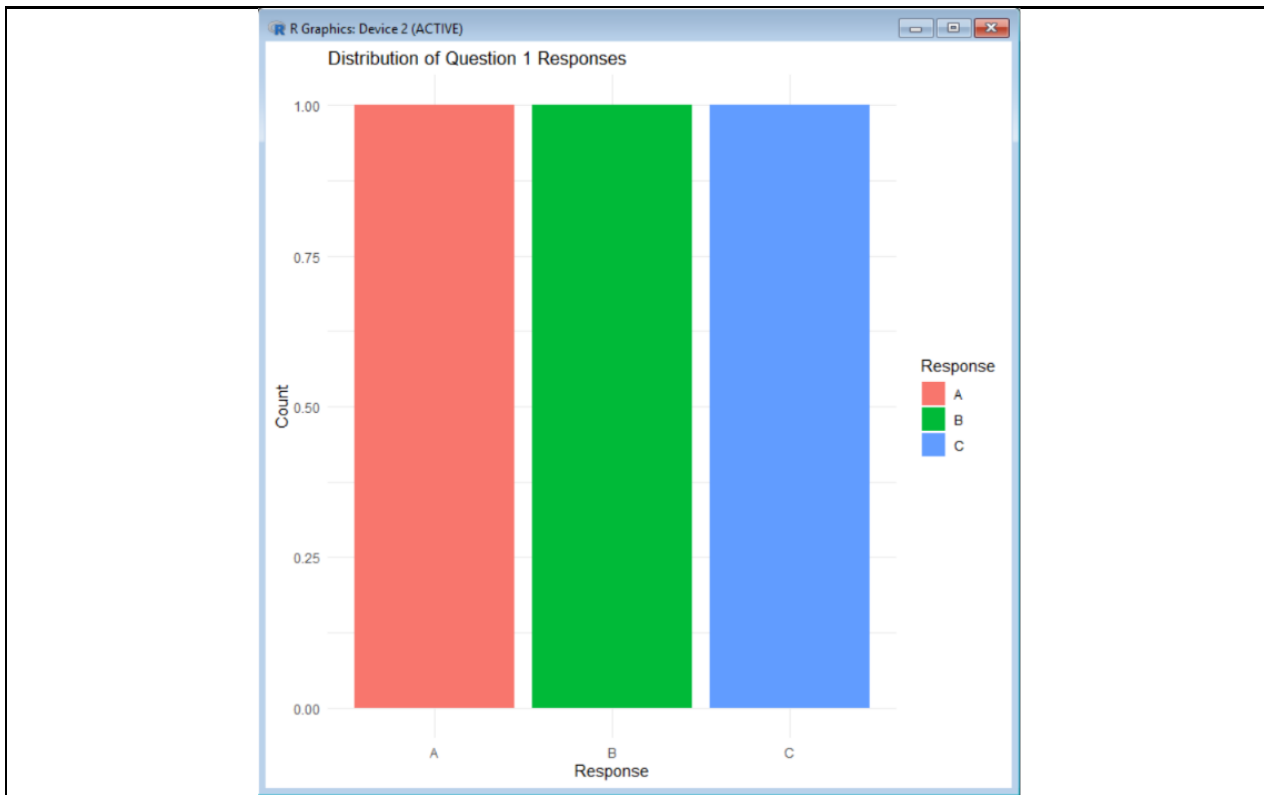
Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Q1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

CODE:

```
library(ggplot2)
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)
q1_data <- as.data.frame(table(survey_data$Question1))
colnames(q1_data) <- c("Response", "Count")
ggplot(q1_data, aes(x = Response, y = Count, fill = Response)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Distribution of Question 1 Responses",
    x = "Response",
    y = "Count"
  ) +
  theme_minimal()
```

OUTPUT:

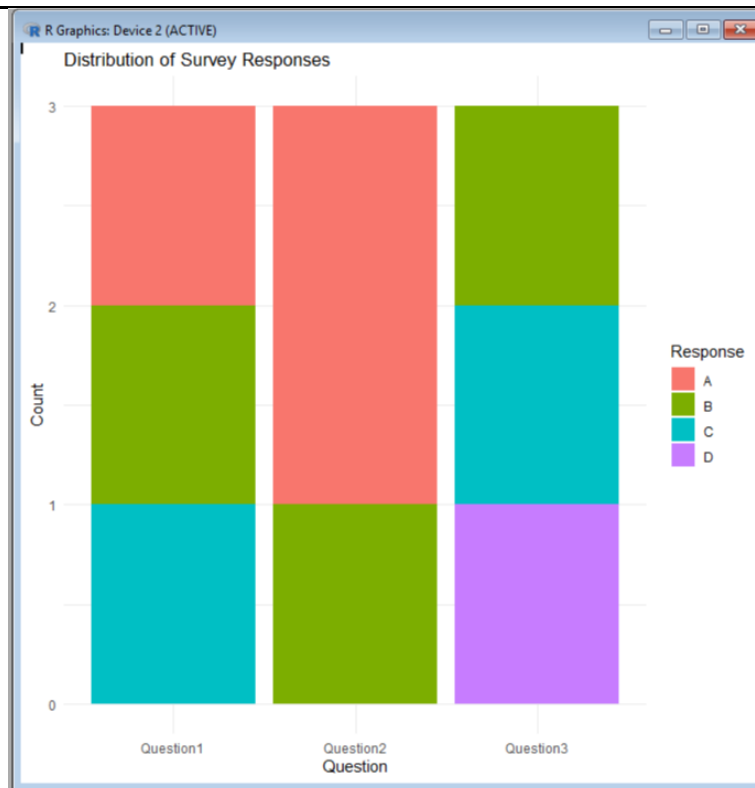


Q2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

CODE:

```
library(ggplot2)
library(tidyr)
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)
survey_long <- pivot_longer(
  survey_data,
  cols = c(Question1, Question2, Question3),
  names_to = "Question",
  values_to = "Response"
)
ggplot(survey_long, aes(x = Question, fill = Response)) +
  geom_bar() +
  labs(
    title = "Distribution of Survey Responses",
    x = "Question",
    y = "Count",
    fill = "Response"
  ) +
  theme_minimal()
```

OUTPUT:



Q3. Build a table to show the survey response data for each survey. Label the table elements.

CODE:

```
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)
print(survey_data)
```

OUTPUT:

Survey_ID	Question1	Question2	Question3
1.00	A	B	C
2.00	B	A	D
3.00	C	A	B

Q4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

CODE:

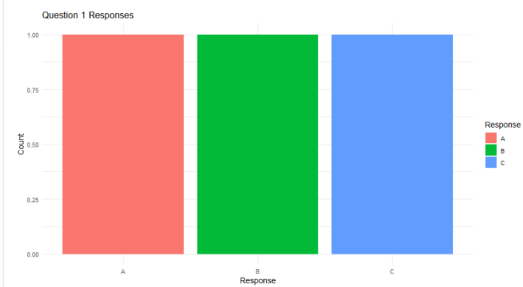
```

library(shiny)
library(ggplot2)
library(tidyr)
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)
q1_data <- as.data.frame(table(survey_data$Question1))
colnames(q1_data) <- c("Response", "Count")
survey_long <- pivot_longer(
  survey_data,
  cols = c(Question1, Question2, Question3),
  names_to = "Question",
  values_to = "Response"
)
ui <- fluidPage(
  titlePanel("Survey Responses Dashboard"),
  fluidRow(
    column(6, plotOutput("barPlot")),
    column(6, plotOutput("stackPlot"))
  ),
  fluidRow(
    column(12, tableOutput("surveyTable"))
  )
)
server <- function(input, output) {
  output$barPlot <- renderPlot({
    ggplot(q1_data, aes(x = Response, y = Count, fill = Response)) +
      geom_bar(stat = "identity") +
      labs(
        title = "Question 1 Responses",
        x = "Response",
        y = "Count"
      ) +
      theme_minimal()
  })
  output$stackPlot <- renderPlot({
    ggplot(survey_long, aes(x = Question, fill = Response)) +
      geom_bar() +
      labs(
        title = "Overall Survey Responses",
        x = "Question",
        y = "Count"
      ) +
      theme_minimal()
  })
  output$surveyTable <- renderTable({
    survey_data
  })
}
shinyApp(ui, server)

```

OUTPUT:

Survey Responses Dashboard



Survey_ID	Question1	Question2	Question3
1.00	A	B	C
2.00	B	A	D
3.00	C	A	B

